Uroš Savurdić

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Serbian and Croatian/EU citizen — no work permit needed in Serbia, the EU/EEA or Switzerland; open to relocation and sponsorship elsewhere.

M.Sc. EECS student and former Applied Science Intern at Microsoft. I build and evaluate LLM systems end to end — retrieval and fine-tuning pipelines, evaluation harnesses, and the measurement work that shows whether any of it actually helps.

EXPERIENCE

Microsoft – Applied Science Intern · LLMs · evaluation · knowledge graphs · multi-hop reasoning — Mar 2026 – Jul 2026

- Built a multi-view evaluation framework pairing a graph-based ranker with a surface-level grader; where the two agree (65.5% of questions) it matched expert difficulty labels 81.2% of the time, against 64.6% for a hop-count policy at the same coverage — a 16.6-point paired gain.
- Developed an end-to-end pipeline generating evidence-grounded, controllable multi-hop QA datasets from enterprise documents using knowledge graphs and LLMs.
- Showed through controlled ablations that a degraded extraction pipeline inflates apparent question difficulty while lowering answerability, so a difficulty metric can reward the wrong thing.

IPSI Belgrade – Research Collaborator · with Prof. Veljko Milutinović — Feb 2026 – present

- Working with Prof. Veljko Milutinović on a comparative survey of emerging computing paradigms for AI workloads.
- Running the literature review and contributing technical review for the survey and an accompanying book.

RESEARCH

Internal Representations of Safety Across Post-Training · Independent research · PyTorch, LoRA, TRL, interpretability — Aug – Sep 2026

- Trained a nine-checkpoint chain on Qwen2.5-1.5B — base → helpful SFT → safety SFT → DPO — crossed over two instruction corpora, with a direct-DPO arm that skips safety SFT as the control. Nine LoRA checkpoints, matched preference data, one factor changed at a time.
- Fixed the benchmark, the direction-estimation procedure and two confirmatory behavioral endpoints in a preregistered analysis plan before looking at any outcome, then recorded every deviation from it.
- Found the harmful/benign contrast forms during instruction tuning rather than DPO (adjacent-stage cosine 0.65 → 0.96 → 0.93), and that DPO amplifies it instead of adding structure: residualize the contrast away and the remaining space carries no linearly decodable harm category, +0.004 with a 95% CI of [-0.018, +0.027].

Attention Mechanisms Across Computational Paradigms · Benchmarking and profiling · PyTorch, JAX, NumPy — Apr – May 2026

- Benchmarked attention across 12 computational paradigms and 17 implementations, decomposed into five stages and measured for FLOPs, memory traffic and arithmetic intensity. Eight implementations ran on hardware I had access to — CPU, an NVIDIA T4 and a TPU v5e; the rest are simulators or analytical cost models.
- Traced a 92x throughput gap in a single PyTorch call to silent backend dispatch between FP32 and FP16, confirmed by logging the selected kernel on every run.
- Built the harness as an extensible package: an abstract paradigm interface, a registry decorator, dataclass metrics and a structured experiment logger.

Presented at the 5th Serbian International Conference on Applied Artificial Intelligence (SICAAI), Kragujevac, May 2026.

PROJECTS

Code Search Engine with Contrastive Learning · PyTorch Lightning, FAISS, Sentence-Transformers — Nov 2025

- Fine-tuned a bi-encoder (all-MiniLM-L6-v2) with InfoNCE contrastive loss on 20k CoSQA query-code pairs, improving MRR@10 from 0.786 to 0.911 (+16.0%), with NDCG@10 +12.1% and Recall@10 +2.8% over the pretrained baseline.
- Enabled semantic search across 550+ indexed code snippets with FAISS and cosine similarity.
- Set up a reproducible training and evaluation pipeline with PyTorch Lightning, AdamW with warmup, Weights & Biases tracking, and a torchmetrics evaluation suite.

Handwritten Text Recognition System · PyTorch, Transformers, Docker, AWS — Aug – Oct 2025

- Built and deployed an end-to-end recognition pipeline: CNN-LSTM and ResNet-Transformer models trained on a modified IAM dataset, with experiment tracking in Weights & Biases, hyperparameter tuning and CI/CD through CircleCI.
- Shipped it as a containerized FastAPI service architected for AWS Lambda.

Applied ML and Data Science Projects · Python, XGBoost, SciPy, LLM APIs — Mar – Oct 2025

- Predicted user engagement to an MAE of 5.06 by engineering 30+ temporal and behavioral features from 90k sessions.
- Evaluated A/B tests using both frequentist and Bayesian methods.
- Automated natural-language queries with intent classification, context extraction, and temporal and geolocation logic.

Optimization Methods · NumPy, from scratch Nov – Dec 2025

- Implemented forward and backward passes by hand in NumPy - manual softmax and ReLU gradients, cross-entropy with log stabilisation - rather than calling an autograd framework.
- Compared gradient descent against a genetic algorithm for training the same network, plotted against wall-clock time so the comparison reflects real cost.
- Implemented 2-opt local search and exhaustive search for a travelling-salesman drilling problem — the heuristic matches the proven optimum on every instance small enough to enumerate, about 70,000x faster at 12 points — plus simulated annealing, a genetic algorithm and differential evolution on constrained problems.

TECHNICAL SKILLS

Languages	Python, SQL, C/C++
ML	PyTorch, Transformers, Hugging Face, LoRA/PEFT, TRL, sentence-transformers, JAX
LLM systems	fine-tuning, DPO, RAG and retrieval, FAISS, evaluation harnesses, LLM-as-judge
Infrastructure	Docker, AWS, FastAPI, Git, CI/CD, Weights & Biases
Spoken	Serbian/Croatian — native · Macedonian — native · English — fluent · Russian and Bulgarian — understood · German — basic

EDUCATION

University of Belgrade — School of Electrical Engineering (ETF) expected Sep 2027

M.Sc. Electrical Engineering & Computer Science · GPA 10/10 · evaluation, system security, optimization

University of Belgrade — School of Electrical Engineering (ETF) graduated Sep 2024

B.Sc. Electrical Engineering & Computer Science · signal processing, information theory

Additional coursework: Full Stack Deep Learning (UC Berkeley); Statistical Learning and CME295 Transformers & LLMs (Stanford).

VOLUNTEERING

EESTEC (Electrical Engineering Students' European Association) Feb 2019 – present

- Delivered 100+ hours of training to 400+ students across Europe and mentored around 30 members into organizational roles; certified Soft Skills Trainer (2023).
- Led a 30-member international team through the COVID-19 transition. Named EESTECer of the Year 2021 across a 4,000+ member network.

AWARDS

EESTECer of the Year — Board of the Association, one recipient per year across a 4,000+ member network (2021) · 3rd place, National Mathematics Competition — Mathematical Society of Serbia (2017) · Best Student Start-Up Award — Science-Technology Park Belgrade (2022)